shepheard.

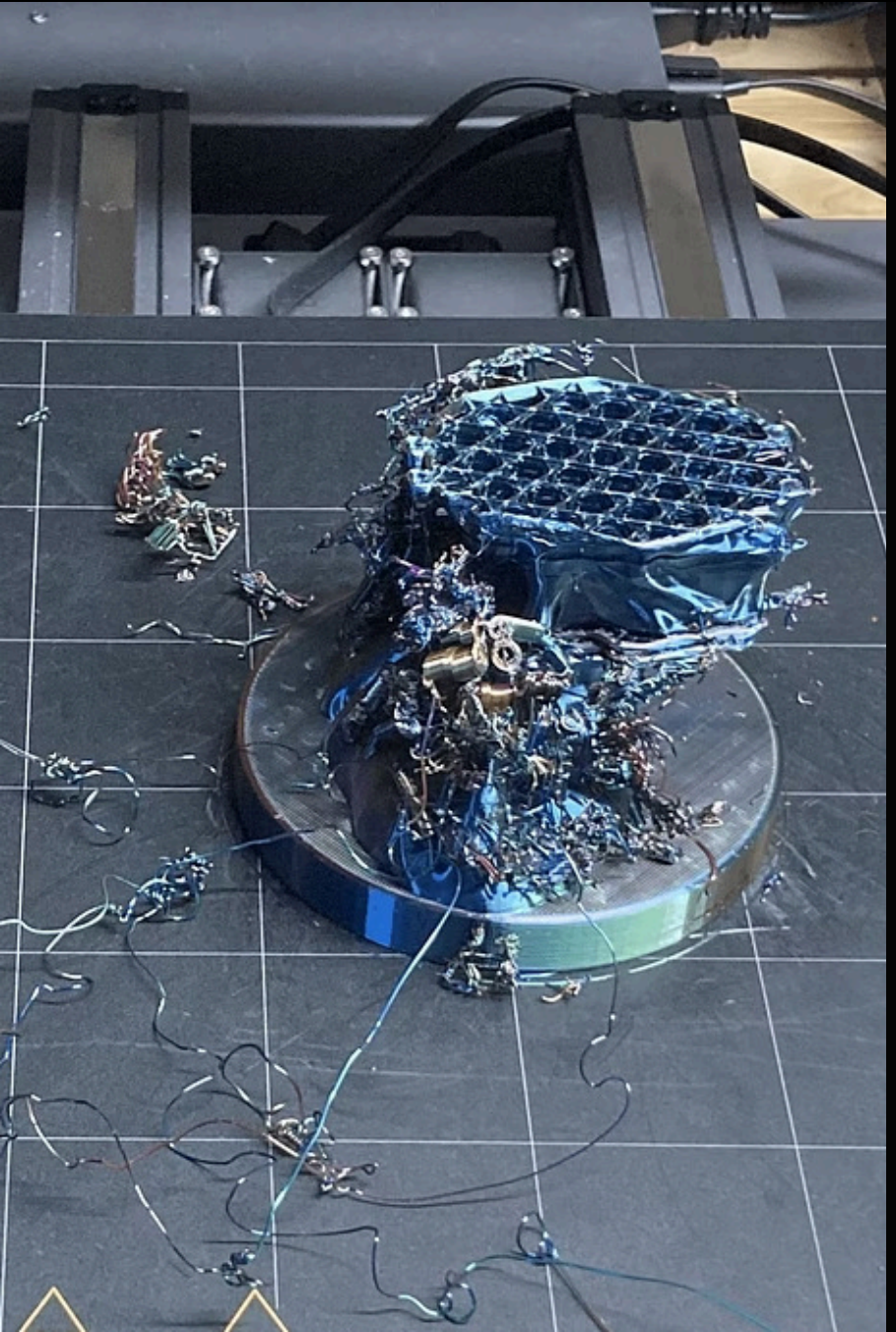
ears for your **machines**

hearing the unheard.

Before we see **danger**, we **hear** it.
Sound is a primal **alarm** - always **on**, even in our sleep.

Yet, the **machines** we build, despite their incredible capabilities, are often **deaf** to their own **plight**.

In each case, the **failure** begins with an **unheard warning**.



the cost of failure: a 3D printing nightmare.

In a **large scale** industrial 3D printing facility, a **critical** aerospace part runs on a **14-hour** print cycle at **95%** capacity.

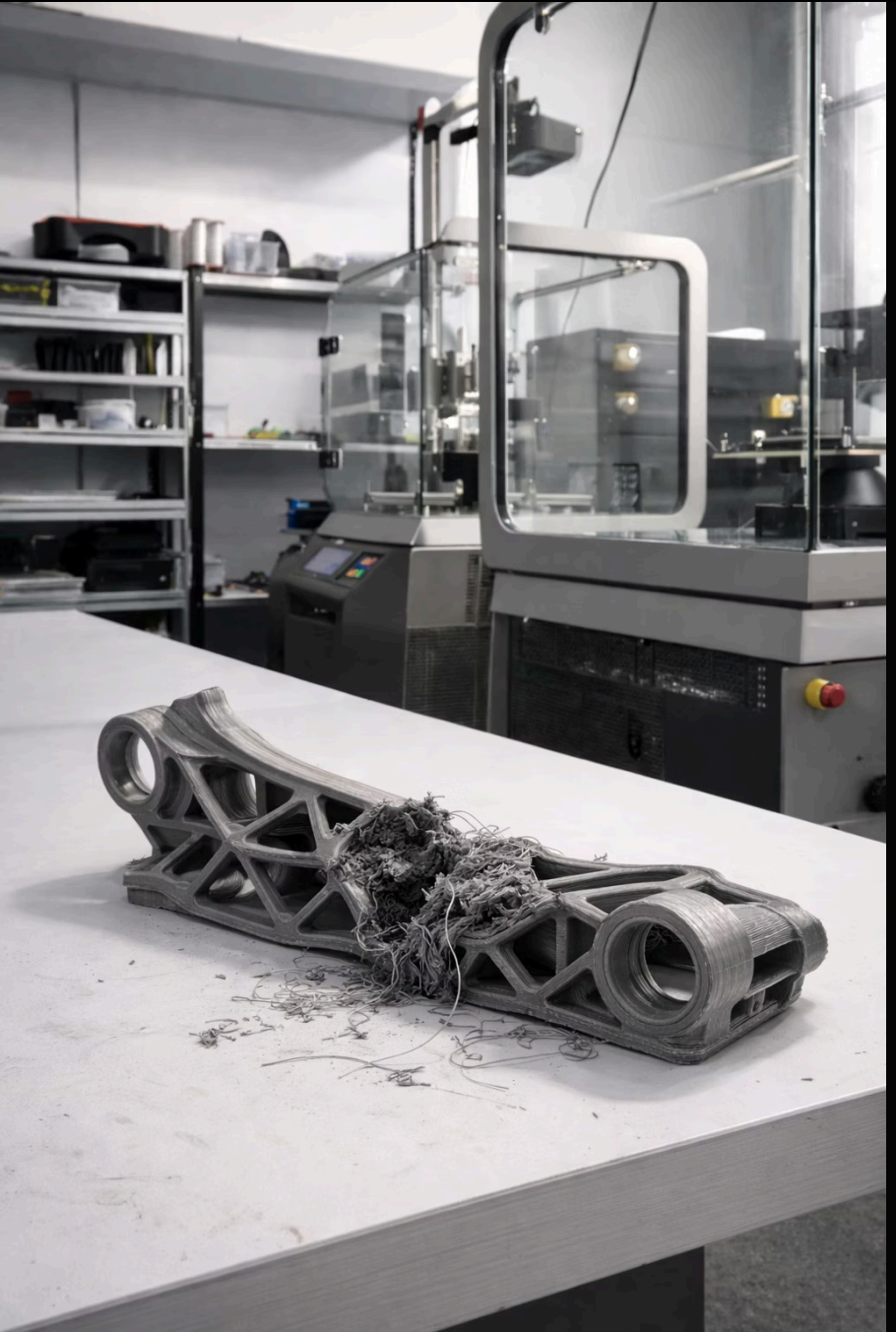
Hours before completion, a **nozzle** begins to **clog**. The warning appears first as a **subtle** change in **sound** and **micro-vibration**—but no **operator** or **sensor** detects it.

The print **fails** catastrophically, wasting both **material** and **14 hours of work**, forcing **unscheduled downtime**, and **missing critical deliveries**. One failure cascades into **lost revenue**, **production delays**, and **damaged reputation**.

shepheard.

We provide a **solution** that can intelligently **analyze** a 3D printer's unhealthy **behaviors** and react **immediately**.

Using **AI powered** audio analysis and **real-time** audio **monitoring**:



Most **failures** are **heard** before they are **seen**

- clogs, wear, slip, and misfeeds change sound first.

Cameras and **sensors** react too **late**

- after geometry or output is already compromised.

Acoustic monitoring is **passive, non-intrusive**, and **scalable**

- enabling early detection across entire facilities.

Making **audio** a powerful and **underused** sensing **layer** across physical infrastructure.

product.

Real-time acoustic **monitoring** detects 3D print **failures** before **damaging** the material and **wasting time**.

- **Microphone array** with **beam steering**
- **AI** audio analysis (real-time)
- Integrated **camera**
- Instant **alerts**

values.

Catch failures **instantly**, not 14 hours later

Save 6-figure losses **per failed print**

Planned **maintenance**, not emergency **shutdowns**

One **operator** monitors **10+ printers**

market.

20.8B

3D Printing Market

The industrial 3D printing market is growing **~15% annually** and expected to reach **USD 20.8 B in 2026**.

125K+

Downtime Cost

Unplanned manufacturing downtime typically costs **~$125,000–$260,000 per hour** in lost production value.

100K+

Single Failure Impact

A single failed long print (e.g., nozzle clog) can cascade into **$100K–$1M+ in losses** through wasted hours, delays, and backlog impact, eroding revenue and customer trust.

gminsights
twi-institute
atlassian

competition.

Squarehead Technologies (Norway)

- Hardware-only, expensive upfront

3DSignals (Israel)

- Proprietary sensors required, high cost

Augury (USA)

- Vibration-only, doesn't work in noisy environments

Arundo Analytics (USA)

- Generic approach, not 3D printing optimized

shepheard. (Latvia)

- AI audio cleaning (software)
- Beam steering (hardware + software)
- Integrated camera (hardware)
- 3D printing optimized (hardware + software)

business model.

SaaS

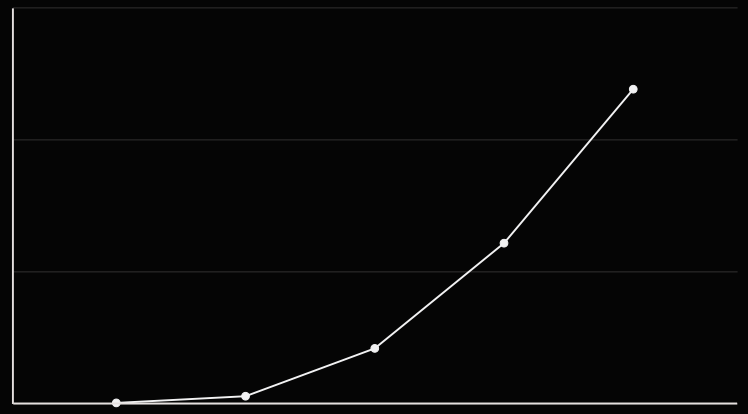
- €50-100/month per printer
- Recurring revenue

Hardware

- Microphone arrays: €200-400
- Camera modules: €150-250
- One-off sales

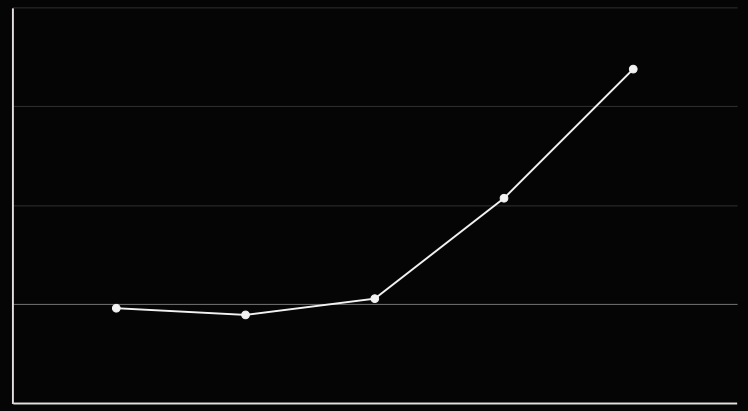
financial projections.

Revenue (€)



Year 1	Year 2	Year 3	Year 4	Year 5
€25K	€320K	€2.5M	€7.3M	€14.3M

Profit Before Tax (€)



Year 1	Year 2	Year 3	Year 4	Year 5
- €170K	- €640K	€70K	€1.5M	€4.3M

go-to-market.

Strategy

1. Target mid-to-large 3D printing shops (50+ printers)
2. 3-month free pilot on 5 printers
3. Prove ROI, then expand to full fleet
4. Aggregate data, improve models, build moat
5. Expand to **CNC**, ship **engines**, other industrial **machinery** and **drone detection**

financial timeline.

YEAR 1 (PRE-SEED)

- Raise pre-seed round
- Goal: create a functioning, testable prototype
- Activities: ongoing real-world testing, interviews, securing local pilot partnerships
- Focus: validate product-market fit and prove technical functionality
- Revenue: minimal (early pilot testing only)

YEAR 2 (SEED ROUND)

- Raise seed round
- Expand team for product development
- Goal: complete pilot integrations with early partners
- Start generating first real recurring revenue
- Reach 20 recurring clients by year-end

YEAR 3–5 (GROWTH PHASE)

- Evaluate need for further investment round
- Full product launch
- Drive recurring revenue through subscription growth (priced at €50-100/mic/month)
- Offer hardware sales (€200-400 per device)
- Emphasize scalable architecture and minimal human deployment
- Head towards 500 recurring clients

team.

Edvards — COO

- 10+ years leading complex audio projects across film, TV, and installations
- Founder of BRODERS AUDIO, managing production teams and clients
- National award-winner for sound design and project execution

Teodors — CTO

- Built audio AI and embedded systems at Neural DSP and MikroTik
- Invented and developed industry-recognized Light Pedal product
- Led academic R&D in spatial audio and neural audio modeling

Krišjānis — CEO

- PE deal support & portfolio company operations at Merito Partners
- Led financial planning and investor materials for SAGA Space Architects deep-tech underwater habitat
- Top student at SSE Riga; Superia award for top finance performance.

the ask.

€200,000 Pre-Seed Round

We are raising **€200,000** to finance **12 months** of operations. This round will fund full-time **work** by the **three co-founders** and allow for **aggressive prototyping** of both **hardware** and **software** components. The goal is to **validate product-market fit** and prepare for scalable deployment.

milestones

- Launch trials with **5 pilot clients**
- Demonstrate **real-time anomaly detection**
- Build foundation for **scalable product rollout**

next round

Next round planned in **12 months**, based on product traction and trial results.

let's give ears to our machines.

We invite visionary investors to join us in shaping the future of industrial reliability. Connect with our founders to learn more about this groundbreaking opportunity.

Edvards — COO

Email: **edvardsbroders@gmail.com**

LinkedIn

Teodors — CTO

Email: **teodors.kerimovs@gmail.com**

LinkedIn

Krišjānis — CEO

Email: **krisjanis.krams@gmail.com**

LinkedIn

appendix: financial projections.

Assumptions	
50%	Cost of hardware
30%	Hardware shipping, storage & other costs
20%	Margin on hardware
75%	Margin on SaaS
€ 75	Subscription per device (monthly)
€ 100	One mic price
€ 200	Camera price
10%	Other ops (tools, customer support etc.) as % of salaries
20%	R&D (product development), 20% of revenue from Y3

appendix: financial projections.

Year	1	2	3	4	5	
Recurring customers	*5*	*20*	*100*	*250*	*500*	
Mics per customer	*5*	*15*	*20*	*25*	*25*	<- industrial level clients need many devices
New mics sold	*25*	*275*	*1700*	*4250*	*6250*	
Mic sales	**€ 2,500**	**€ 27,500**	**€ 170,000**	**€ 425,000**	**€ 625,000**	
Camera per customer	*0*	*1*	*5*	*5*	*5*	
New cameras sold	*0*	*20*	*480*	*750*	*1250*	
Camera sales	**€ -**	**€ 4,000**	**€ 96,000**	**€ 150,000**	**€ 250,000**	
Subscription revenue	**€ 22,500**	**€ 288,000**	**€ 2,250,000**	**€ 6,750,000**	**€ 13,500,000**	
Total revenue	**€ 25,000**	**€ 319,500**	**€ 2,516,000**	**€ 7,325,000**	**€ 14,375,000**	
Hardware COGS	**€ 2,000**	**€ 25,200**	**€ 212,800**	**€ 460,000**	**€ 700,000**	
Saas COGS	**€ 5,625**	**€ 72,000**	**€ 562,500**	**€ 1,687,500**	**€ 3,375,000**	
COGS	**€ 7,625**	**€ 97,200**	**€ 775,300**	**€ 2,147,500**	**€ 4,075,000**	
Gross margin	*70%*	*70%*	*69%*	*71%*	*72%*	
Nr. Of employees (Y1 only founders)	*3*	*8*	*15*	*30*	*45*	
Average monthly salary of new employees (incl. LV tax)	*€ 2,500*	*€ 6,500*	*€ 5,500*	*€ 4,500*	*€ 4,000*	<- employees from Y2 must be top-tier for a strong development and expansion team
Average annual salary	€ 30,000	€ 60,000	€ 62,800	€ 58,400	€ 54,933	
Salaries	**€ 90,000**	**€ 480,000**	**€ 942,000**	**€ 1,752,000**	**€ 2,472,000**	
Sales & marketing per customer	*€ 2,000*	*€ 1,600*	*€ 1,280*	*€ 1,024*	*€ 819*	<- learning economies set at 80% per year
Sales & marketing	**€ 10,000**	**€ 32,000**	**€ 128,000**	**€ 256,000**	**€ 409,600**	
Other ops (tools, customer support etc.)	**€ 9,000**	**€ 48,000**	**€ 94,200**	**€ 175,200**	**€ 247,200**	
R&D (product development)	**€ 75,000**	**€ 300,000**	**€ 503,200**	**€ 1,465,000**	**€ 2,875,000**	
OPEX	**€ 184,000**	**€ 860,000**	**€ 1,667,400**	**€ 3,648,200**	**€ 6,003,800**	
Operating margin	*-667%*	*-200%*	*3%*	*21%*	*30%*	
Total expenses	**€ 191,625**	**€ 957,200**	**€ 2,442,700**	**€ 5,795,700**	**€ 10,078,800**	
PBT	**€ (166,625)**	**€ (637,700)**	**€ 73,300**	**€ 1,529,300**	**€ 4,296,200**	<- no income tax, let profit=retained cash
Funding year-start	€ 200,000	€ 1,000,000	€ 2,000,000	€ -	€ -	<- 3rd year funding likely needed as cash cycle won't be fast enough yet
Cash year-start	€ 200,000	€ 1,033,375	€ 2,395,675	€ 2,468,975	€ 3,998,275	
Cash year-end	€ 33,375	€ 395,675	€ 2,468,975	€ 3,998,275	€ 8,294,475	